TEAM THERMOCLINE

Preliminary Test Readiness Review

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Project Team & Roles

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System Engineer
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Agenda

- System Description
- Testing: Electronics Subsystem
  - Overview
  - Load Driving
  - Current Monitoring
  - Temperature and Analog
  - Misc and I/O
  - Future Testing
- Testing: Mechanical Subsystem
  - CAD Review
  - "CAD"
  - Frame Assembly
  - Future Testing
- Testing: Thermo Fluid
  - Bottleneck
  - Future Testing
- Design Changes
- Questions

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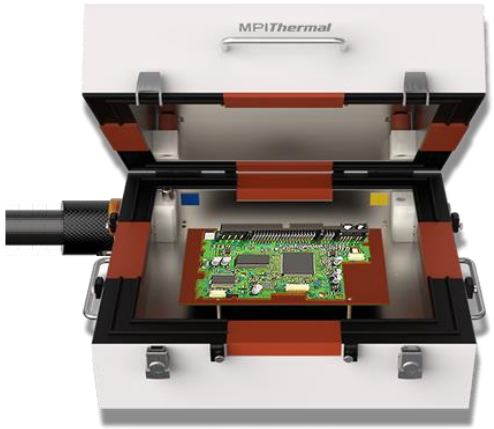
System Description

- This system will be a thermally controlled Printed Circuit Board ("PCB") testing chamber. It is a **table-top** system designed to be easy to use and operate.
- It will test and evaluate PCBs through changing or static temperature ranges, a common method of PCB evaluation



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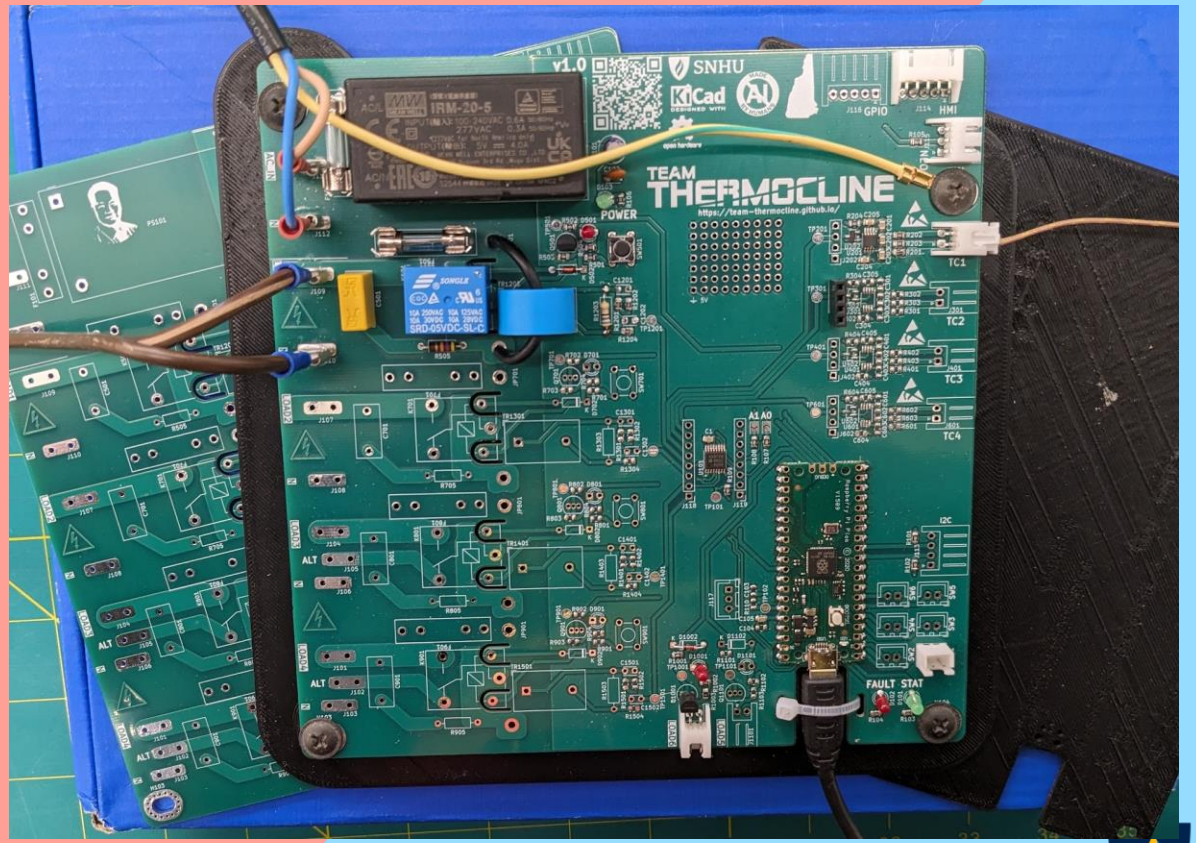
Testing: Electronics Subsystem



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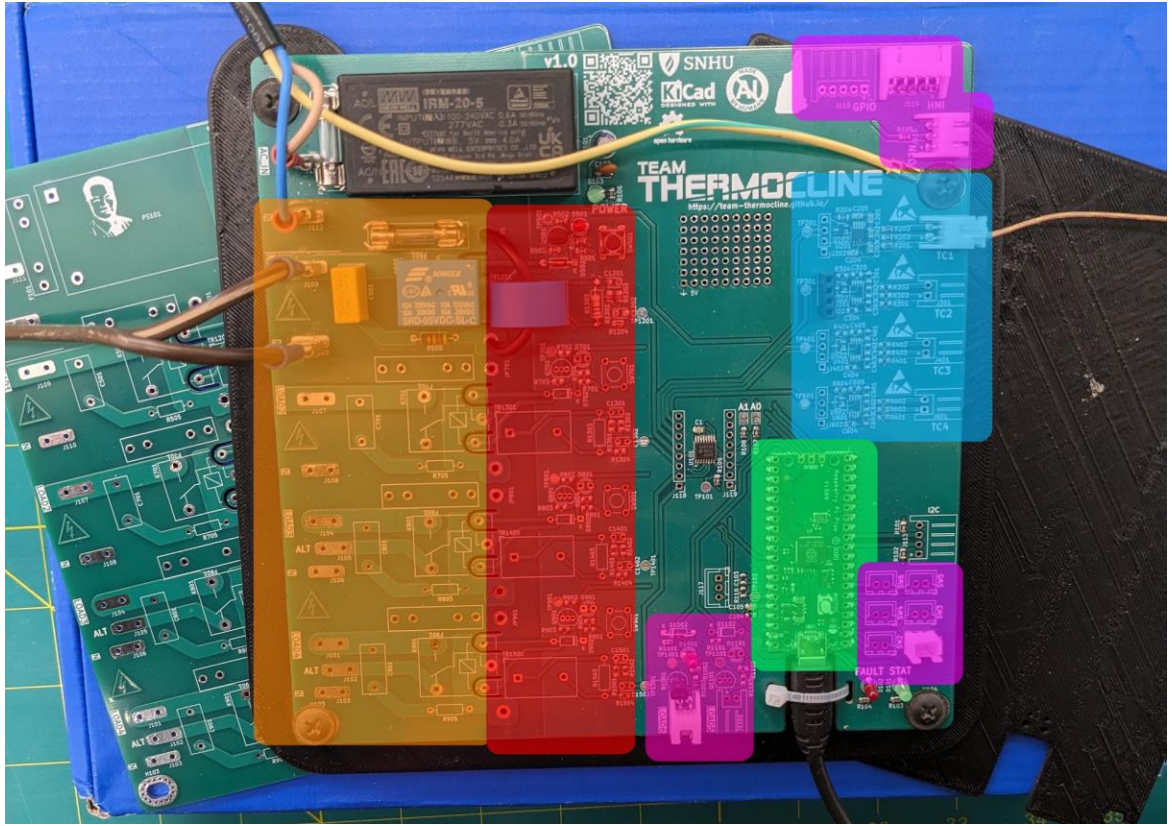
Overview



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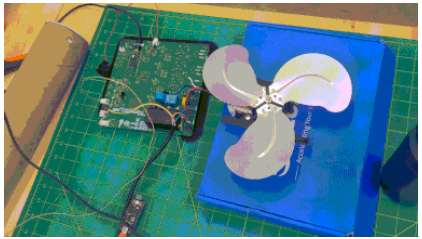
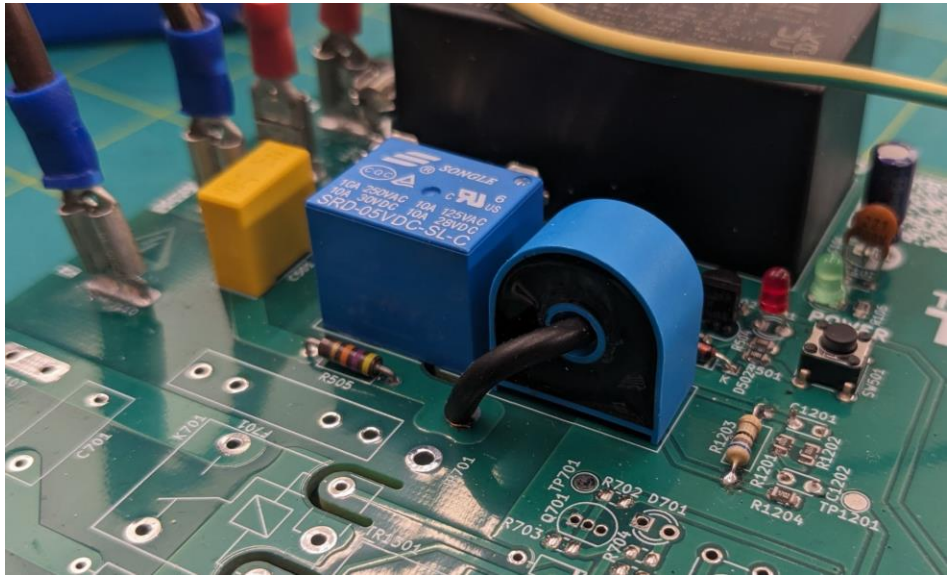
Load Driving

- Many parts of the system rely on power switching
- Each subsystem has their own power requirements
- Test procedures cover the extremes of all values





Metric	Result	Success Criteria
Min. Switching Time	<100ms	<100ms
Maximum Load (Cont.)	>15A	>10A
Snubbed EMI	None (As measured on ADC)	None
Controls & Manual	Functional	Functional

EMI: Electromagnetic Interference, ADC: Analog to Digital Converter

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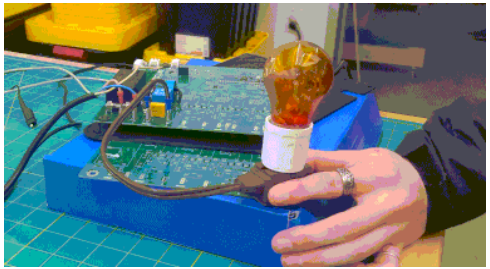
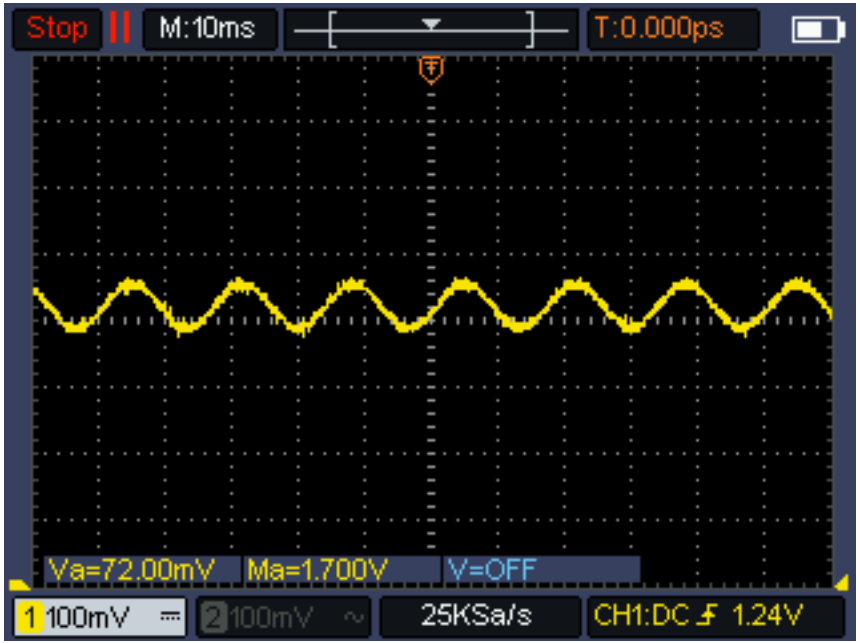
Current Monitoring

- Load monitoring affects system safety
- Allows the CPU to better monitor the state of the cooling system
- Cooling systems are expensive and delicate, dedicating hardware to monitor the performance protects the system long term





Metric	Result	Success Criteria
Amplitude Limit	2.7V	<3.3V
Symmetrical Sinusoid	Yes (up to 25A)	>10A
ADC tracking	Yes	Yes
Calibration	1.1% Error	<10% error

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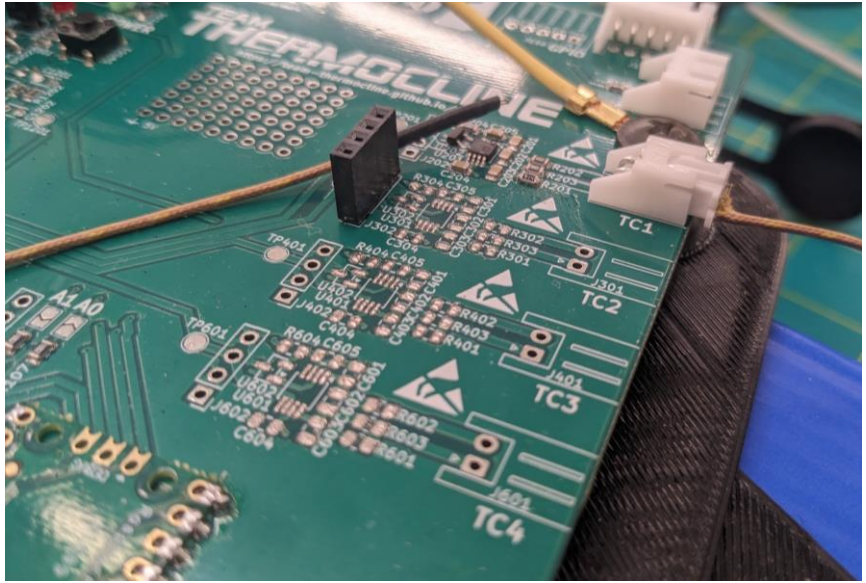
Temperature and Analog

- The system involves a many thermocouple sensors to monitor many conditions.
- The sensors must be reliable and tested at range before use.
- The sensors must be calibrated and agree with each other and agree with external tools.



Metric	Result	Success Criteria
Agrees with each other	<0.5°C	<1.5°C
Crushed Ice Test	0°C	±1.5°C
ADC tracking	>500°C	>90°C
Update Frequency	>100Hz	>10Hz

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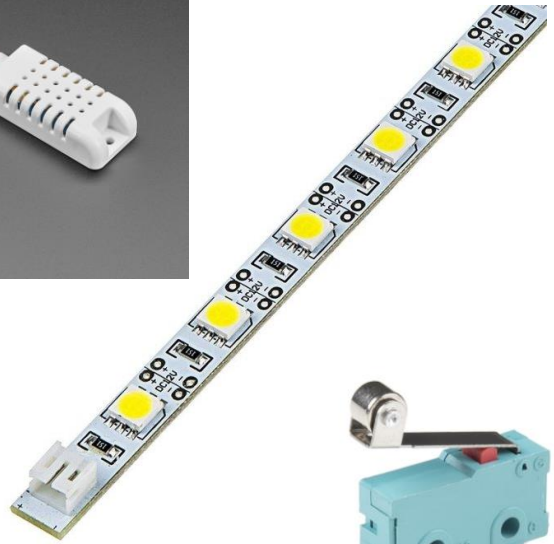
Misc. And I/O

- The system has more inputs and outputs than just power and analog.
- Some sensors and drivers need digital or data I/O
- Inter-module communication is required for a distributed system



Metric	Result	Success Criteria
HMI Serial	Functional	Functional
Neopixel Output	Functional	Functional
Safety and DIO switches	Functional	Functional
I2C intercommunication	Functional	Functional

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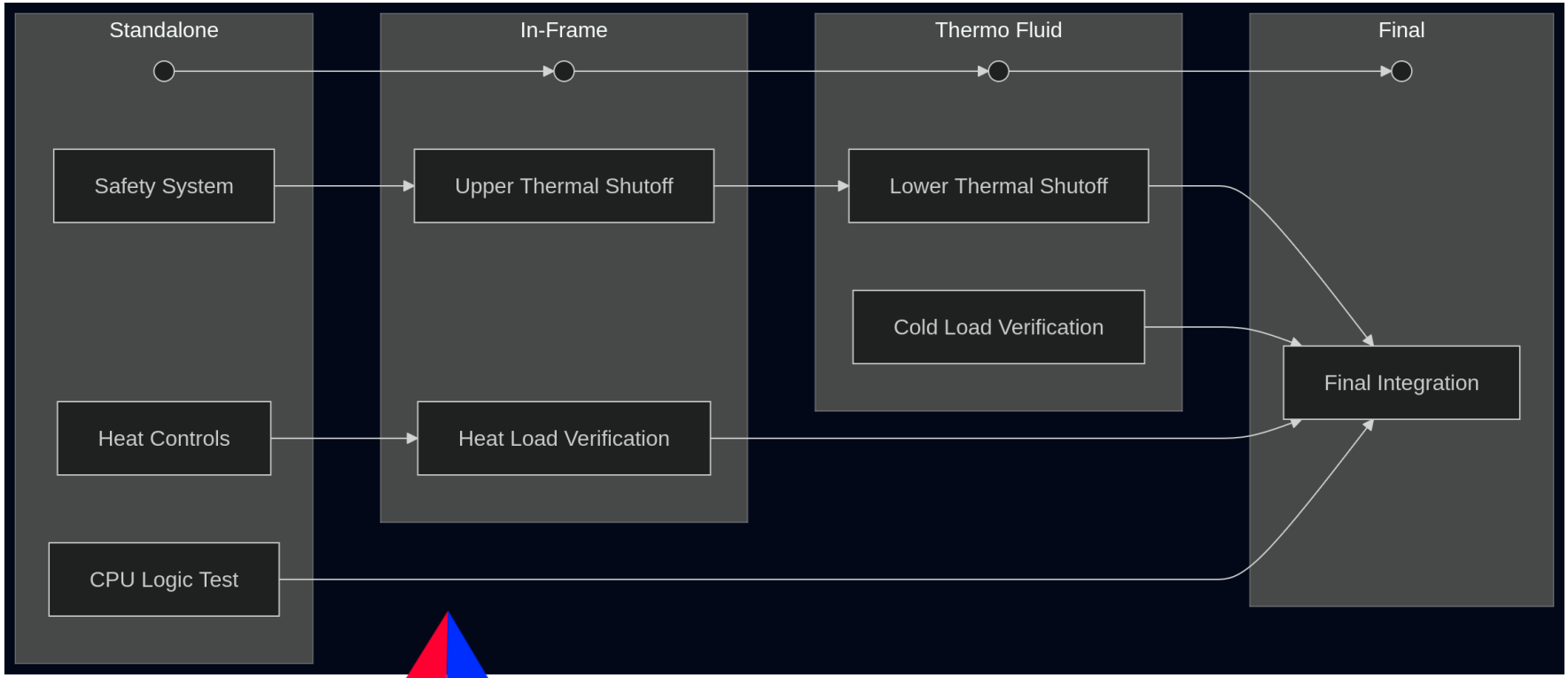
Future Test Plan (Electrical)



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Plan to complete
All "In-Frame" tests before March 10

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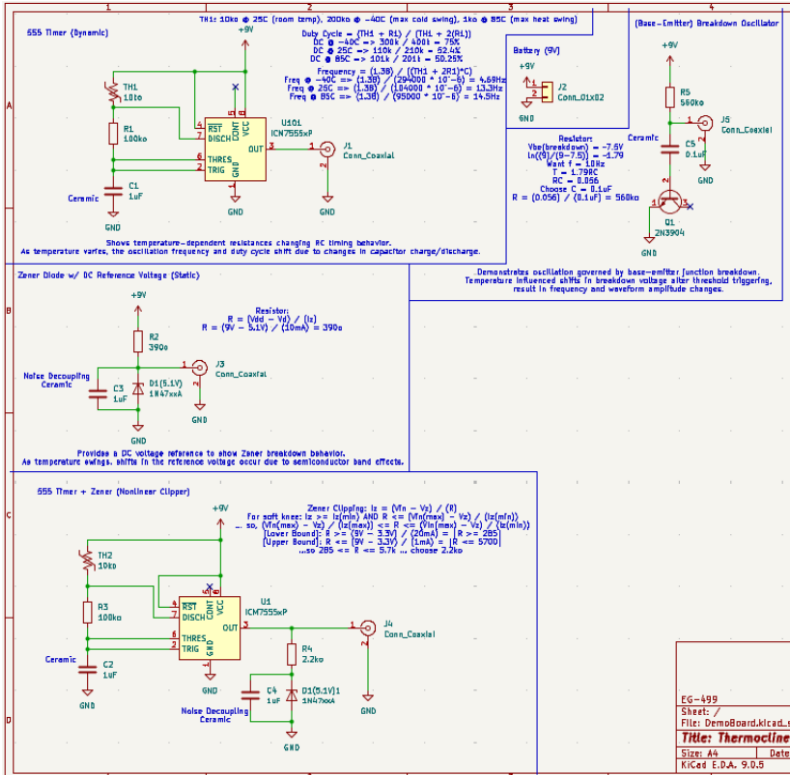
Future Tests



Test Board Schematic to be used to verify effects of chamber on future customer PCB

Overcurrent and Fault System

Required Tools: Current Probe, Voltmeter

Test Procedure: Purposefully cause (safely) one of every major fault to ensure that the computer and electrical system respond and handle it. Those most important being: Chip detection, Bus Fault, Overcurrent Condition, Overheat, Stalled heat or Stalled Cooling and Watchdog Fault

Post Op: Every fault must be handled either in hardware or software and must be repeatable. Faults must present themselves through a fault indication in software or automatic shutdown in hardware.

Safety and Personnel: Two or more team members must conduct the experiments for safety

Success Criteria: No common fault conditions are left un-handled.

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Future Tests

Upper and Lower Limit Tests

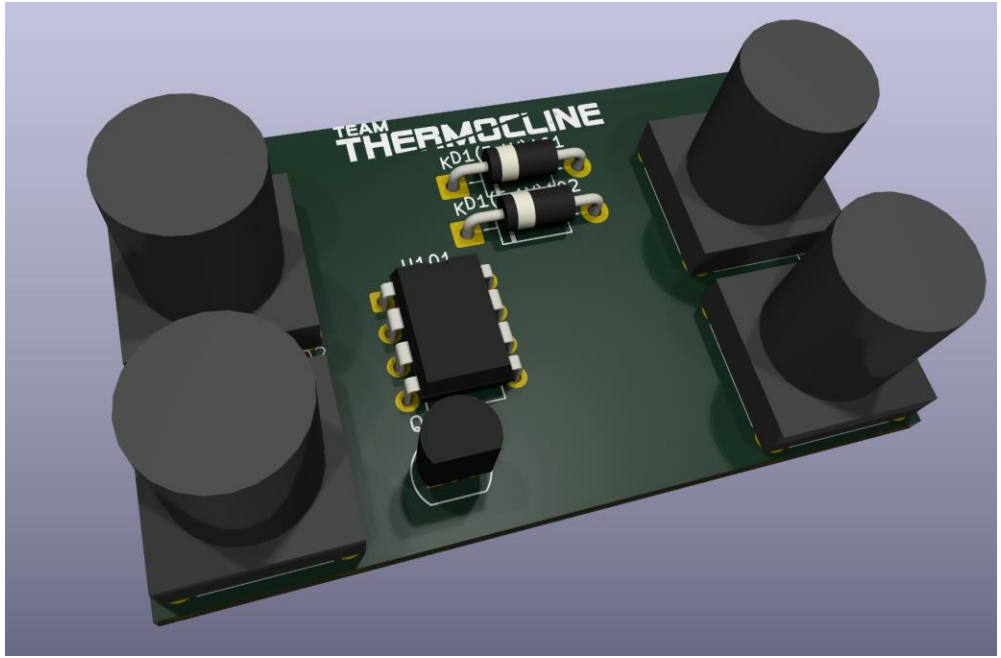
Required Tools: N/A

Test Procedure: System will heat to the upper limit and remain there to saturate all thermal mass with heat, then drive to cool and remove the heat once saturation occurs.

Post Op: We must carefully examine the load placed on the compressor at its peak and most extreme load to ensure it is not overloaded, if we find it to be overloaded, sizing adjustments must be made.

Safety and Personnel: An observer to monitor the computer and ensure load data is saved and that the system never overloads.

Success Criteria: Complete function from hot to cold without overloads.



Test Board to be used to verify effects of chamber on future customer PCB

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Future Tests

HMI and User Interface

Required Tools: Laptop

Test Procedure: The embedded HMI must perform to the same standard as the standalone HMI and must have feature parity.

Post Op: If any features are incomplete, missing or improper implementation they must be corrected

Safety and Personnel: One team member will use the simulator or real controller and verify each function on the UI.

Success Criteria: No required features as defined in the requirements database will be incomplete, missing or broken.

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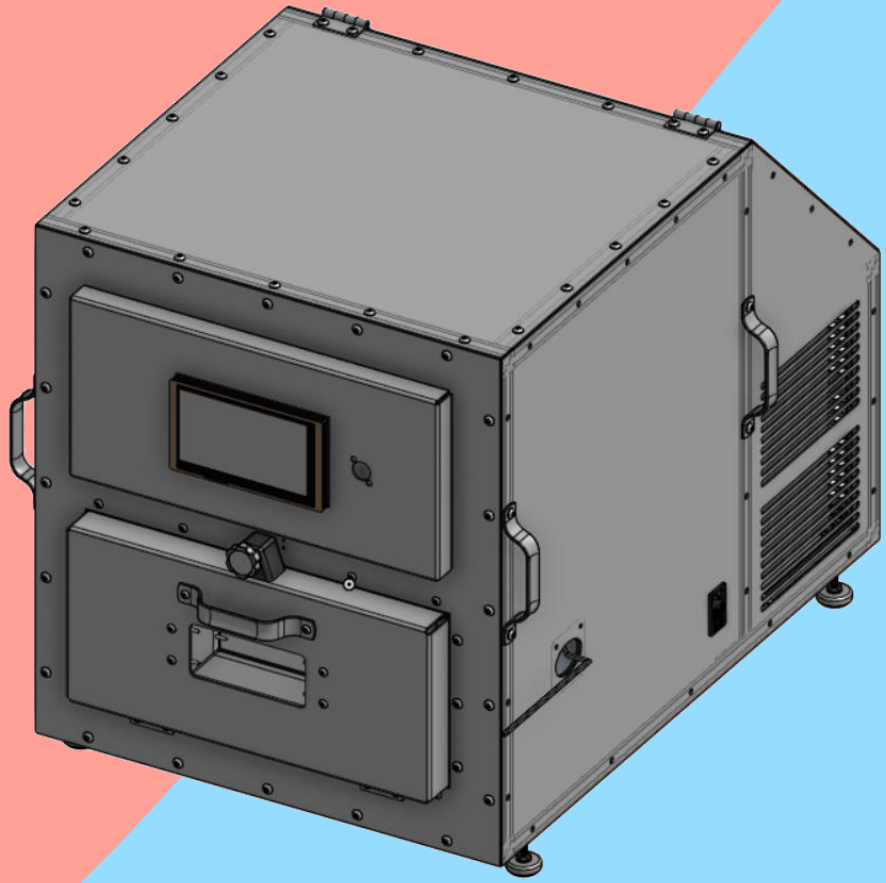
Testing: Mechanical Subsystem



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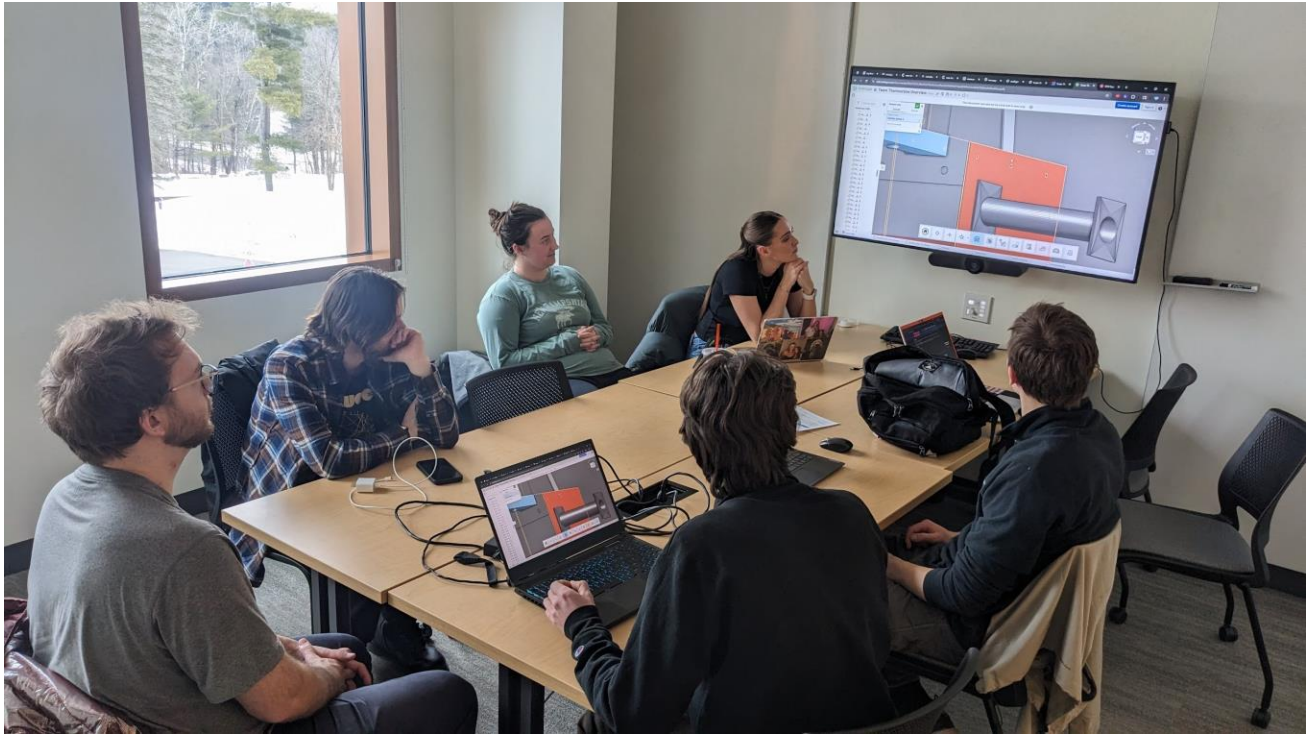
CAD Review

- CAD review completed on 2/17
- Many notes were taken on how to verify design and what to test post-production



Thank you everyone who gave feedback!

John Vincent
Professor Monk
Ruben Del Rio Ruiz

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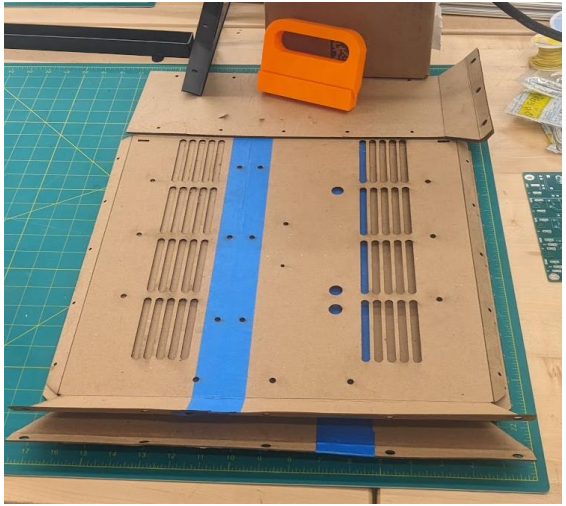
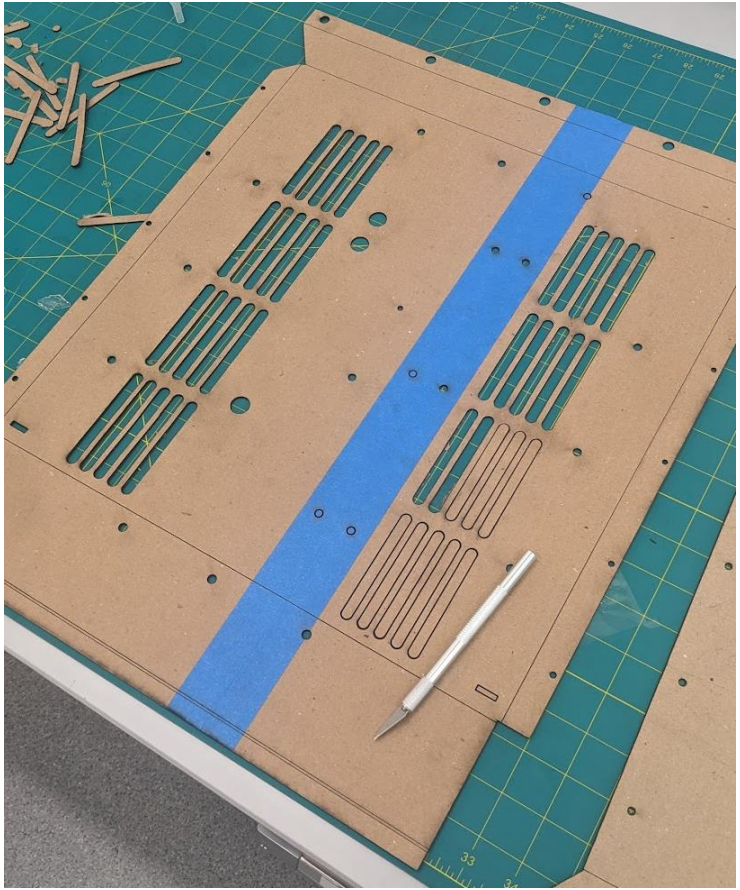
"CAD" Verification

- **C**ardboard **A**ided **D**esign
- Many parts could be tested for fit and function (sans temperature swings)
- Success Criteria for fitment checks: "If it fits, it ships!"
- Tolerances account for future manufacturing capabilities





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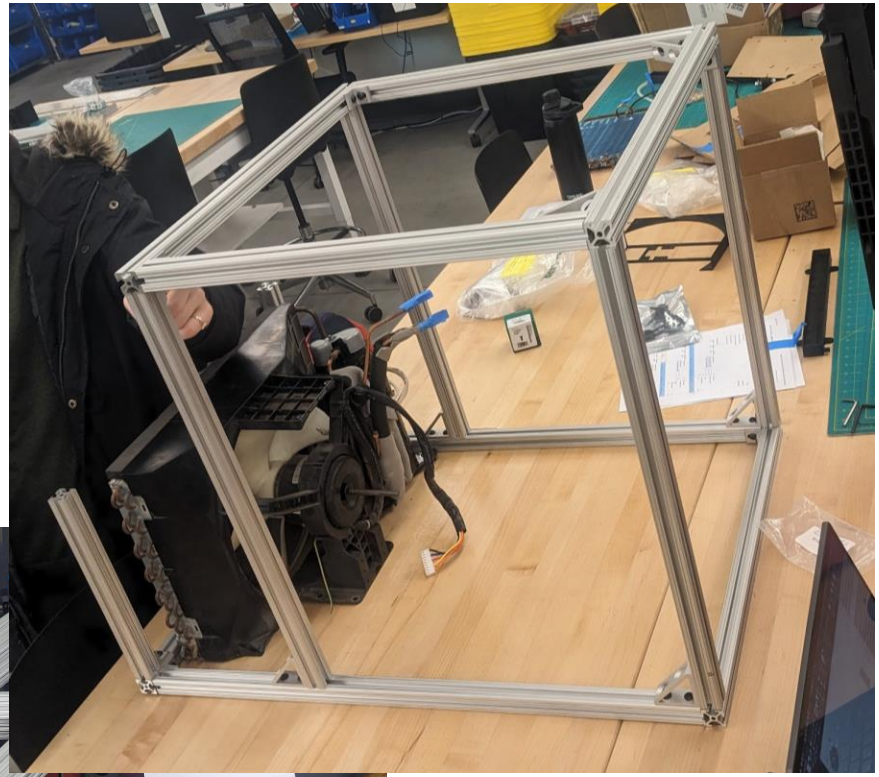
Frame Assembly

- Frame assembly gives us references to measure our manufactured parts off of
- Frame measurements all within our 1/32" spec
- Known frame of reference to work from



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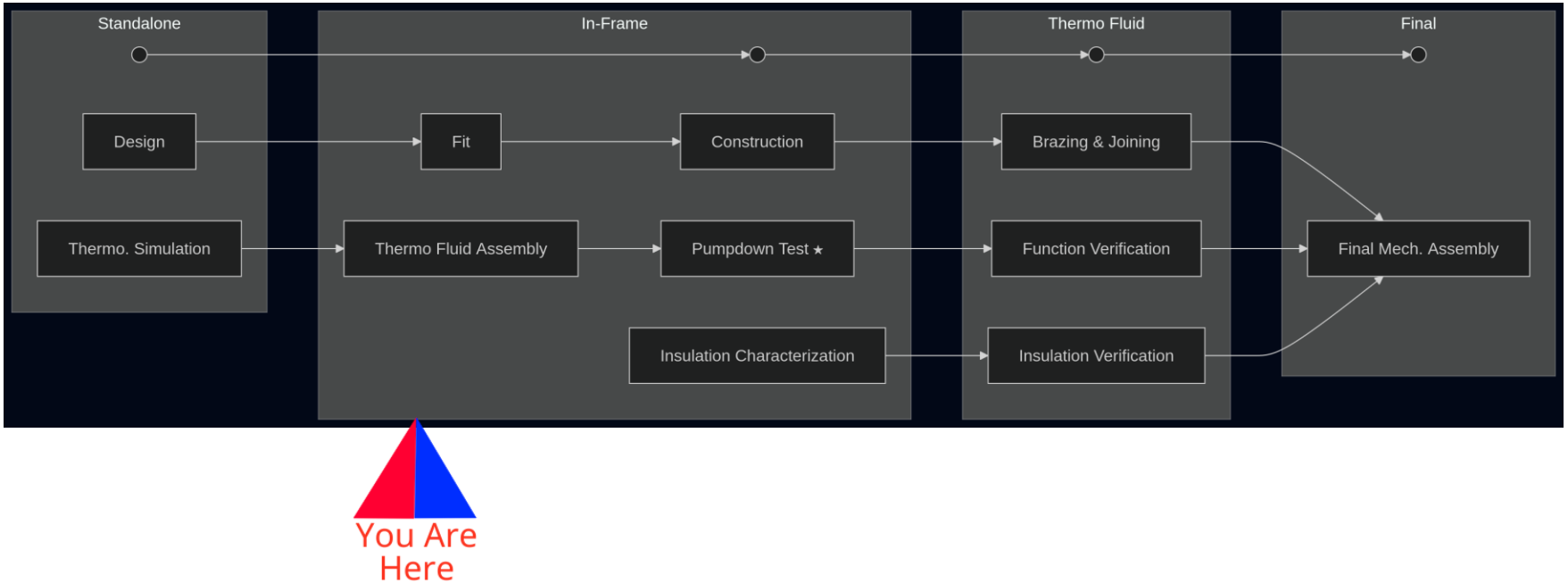
Future Test Plan (Mechanical)



Plan to complete
All "In-Frame" tests before March 17

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Future Tests: M1

Mechanical Body Integrity

Required Tools: N/A

Test Procedure:

1. Two people lift up the non-operating system, holding it for 5-10s, and gently placing it down.
2. Body is examined for weak points or failed joints/fasteners.
3. Procedure performed again while system is running.

Post Op: If system fails, body is strengthened, certain parts are redesigned, or different components are selected/ordered.

Safety and Personnel: Lifting system (2 people), Observer/Examiner (1 person).

Success Criteria: Entire system exhibits no weak points, remaining secure throughout tests and beyond.

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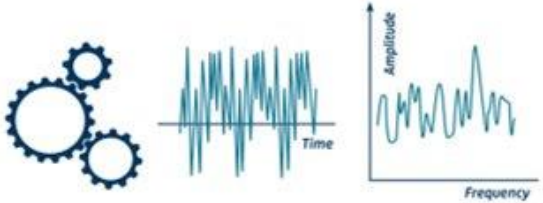
Future Tests: M2

- **Vibrational Analysis**
- **Required Tools:** Phone (vibrational accelerometer application)
- **Test Procedure:**
- 1. Securely attach phone to a part of the external frame.
- 2. Engage/disengage entire system.
- 3. Analyze vibrational data.
- 4. Repeat procedure on other critical parts of the frame, and other main internal components – with higher damage risk.
- **Post Op:** If components or regions of system experience any damage, modify fasteners/joints, redesign parts of system, build or order dampers/padding.
- **Safety and Personnel:** Vibrational measurer (one person), Examiner(one person).
- **Success Criteria:** Subsystems show no areas prone to long-lasting wear/damage due to vibrational impacts.



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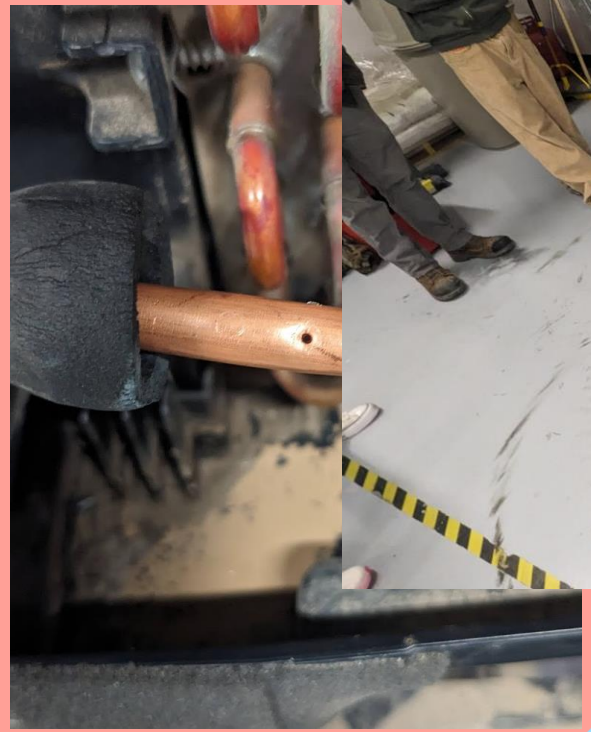
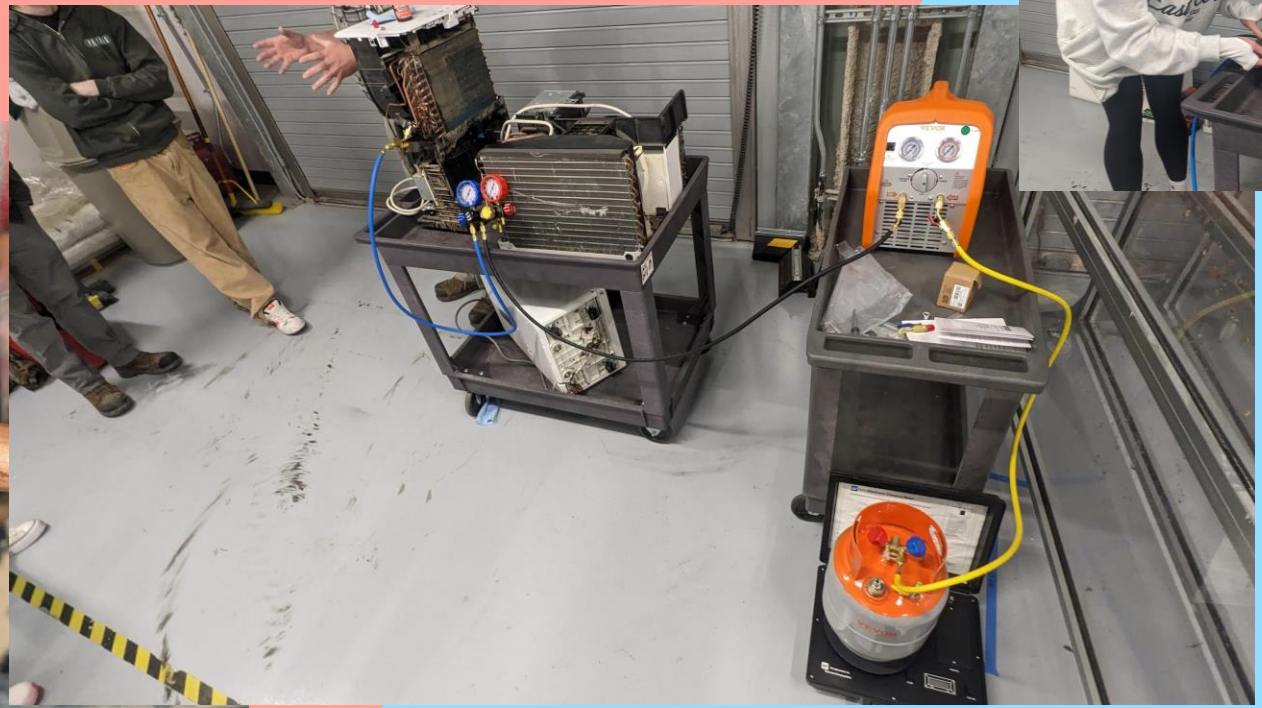
Thermo Fluid (And Next Steps)





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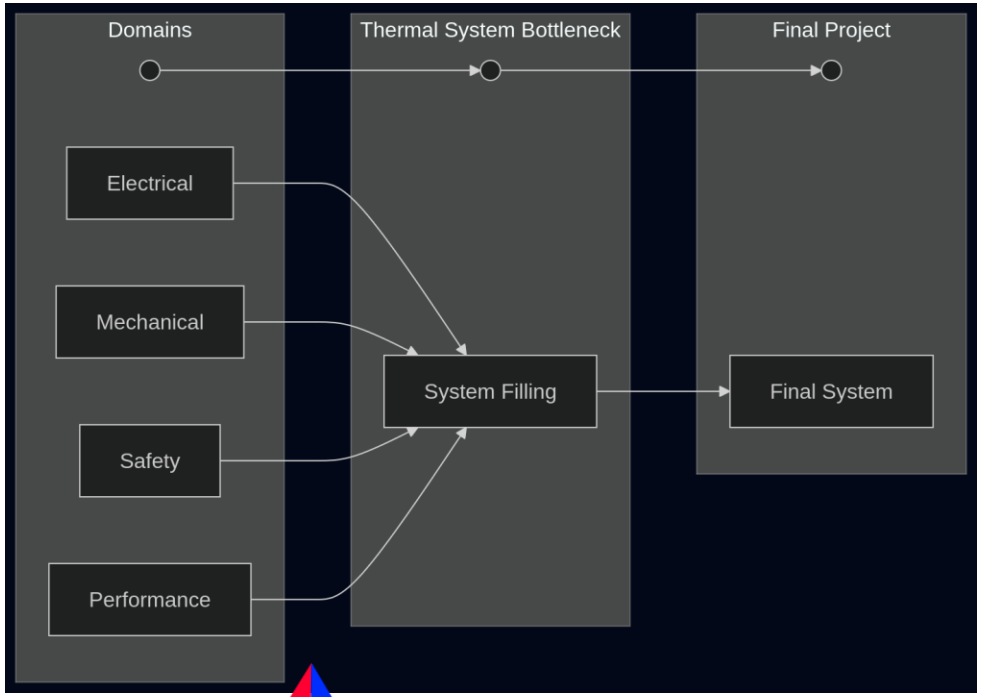
Thermal Fluid Bottleneck



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Many important tests are forced to wait until we can refill the system with refrigerant

Tests must take into account limited time post-installation

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Future Tests: Th1

Fluid Characterization

Required Tools: Pressure Gauges, External Probes, Oscilloscope, Clamp Meter, Laptop

Test Procedure: System will run compressor and operators monitoring current consumption, thermal fluid pressure differential and evaporator point to tune in flow rate, fluid volume inside the system, gas expansion and generally draw out the system to fine tune its performance.

Post Op: Performance must be verified after sitting to ensure system is not overfilled, underfilled or improperly governed.

Safety and Personnel: All hands on deck, everyone will be needed to ensure safety and operate all the various tools at once. Make calculations, monitor, ensure safety.

Success Criteria: System will perform to performance requirements.

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Future Tests: Th2

- **System Heat Transfer**
- **Required Tools:** IR Thermometer
- **Test Procedure:**
- 1. Engage fan & thermal-fluid system (at different power levels)
- 2. Measure temperature points (electrical power conversion, air, insulation, external environment)
- 3. Analyze differential
- **Post Op:** Modify system (insulation, seals, fan speed, etc.) as needed if certain temperature areas are not as desired.
- **Safety and Personnel:** Operator/controller (one person), thermal temperature data capturer (one person)
- **Success Criteria:** System will be properly sealed according to functional requirements.

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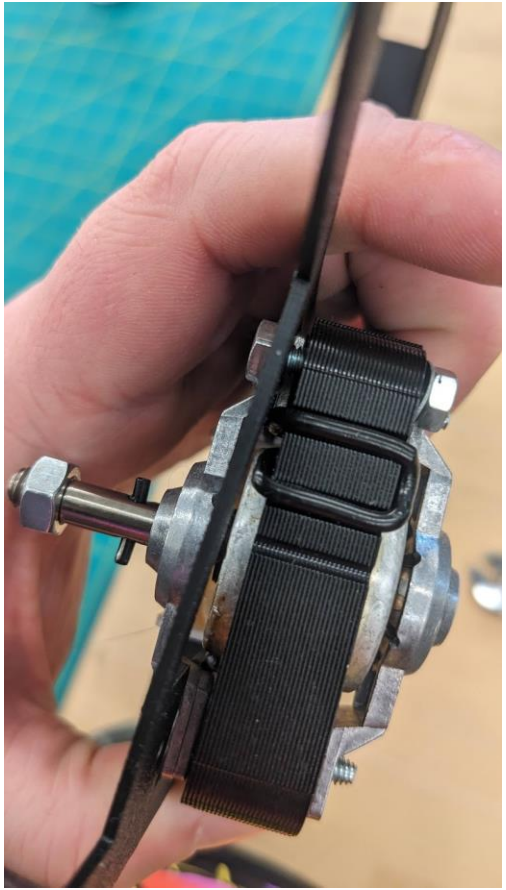
Design Changes

- Door panel was simplified and broken into sub-parts in some places
- Inserts for thermal components
- Dimensions tweaked for ease of manufacturing
- HMI Screen swapped out for an equivalent



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Project Budget

To date, we have spent **$1,109.51**

This represents **36.98%** of our total budget.

Not including tools, we have spent **$606.51** which represents **20.22%**

We have **$1,890.49** remaining to spend in total, **$1,290.49** factoring in allocations for the management budget

We plan to spend **$1,000** for sheet metal parts, or **78%** of our total remaining budget.

Cost	Link	Date	Sybsystem		Total	% of budget	Left to spend
$ 35.00		?	Electrical		$ 1,109.51	36.98%	$ 1,890.49
$ 30.00		?	Mechanical				
$ 50.00		?	Electrical				
$ 388.00		?	Tools		Total (Not Including tools)		
$ 49.00		?	Tools		$ 606.51	20.22%	
$ 66.00		?	Tools				
$ 25.55		?	Electrical				
$ 23.17		?	Electrical				
$ 12.69		?	Mechanical				
$ 83.54		Friday, January 16, 2026	Electrical				
$ 96.00	link	Friday, January 30, 2026	Electrical				
		Tuesday, January 27, 2026	Mechanical				
$ 14.99	Grove Link	Friday, February 13, 2026	Electrical				
$ 14.99	Amazon	Friday, February 13, 2026	Electrical				
$ 65.95	Canakit	Friday, February 13, 2026	Electrical				
$ 66.79	Amazon	Friday, February 13, 2026	Electrical				
$ 16.91	Amazon	Friday, February 13, 2026	Mechanical				
$ 13.99	Amazon	Friday, February 13, 2026	Electrical				
$ 9.99	Amazon	Friday, February 13, 2026	Electrical				
$ 5.99	Amazon	Friday, February 13, 2026	Electrical				
$ 5.99	Amazon	Friday, February 13, 2026	Electrical				
$ 13.99	Amazon	Friday, February 13, 2026	Electrical				
$ 12.99	Amazon	Friday, February 13, 2026	Electrical				
$ 7.99	Amazon	Friday, February 13, 2026	Electrical				

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Questions?

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Backup Slides

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Requirements Database- Functional

3.1	**The system shall monitor temperature inside the chamber.**	Demonstration
3.1.1	The sensors shall be properly rated to withstand internal environmental conditions.	Analysis
3.2	**The system shall control temperature inside the chamber.**	Demonstration
3.2.1	The system shall cover *commercial* temperature range (0oC - 70oC). [Optional]: cover *industrial* temperature range (-40oC - 85oC).	Measure
3.2.2	The system shall allow local temperature setpoint control.	Test
3.2.3	The system shall allow remote temperature setpoint control.	Test
3.2.4	The system shall maintain a toleranced temperature at ±3°C of setpoint.	Measure
3.2.5	The system shall prohibit thermal control when the chamber door is open.	Demonstration
3.2.6	The system shall prohibit thermal control when a user is handling the PCB.	Demonstration
3.2.7	The system shall include a safety measure if temperature range & tolerance is exceeded.	Demonstration
3.3	**The system shall remain within a specified humidity range (0% - 90%).**	Measure
3.4	**The system shall provide data output.**	Demonstration
3.4.1	The system shall support PC connection for datalogging.	Inspection
3.4.2	The system shall log temperature data.	Test
3.4.3	The system shall log humidity data.	Test
3.4.4	The system shall log faults for safety measures.	Measure
3.4.5	The system shall be capable of sweeping all temperature ranges within a 24-hr cycle.	Test

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Driving Requirements

- **1.1** - The system shall accommodate a 12x12" PCB as a device to be tested.
- **2.1.1** - The PCB shall successfully operate while temperature is fluctuating.
- **3.2.1** - The system shall cover commercial temperature range (0-70°C)
- **3.2.4** - The system shall maintain a temperature ±3°C of setpoint.
- **3.4.5** - The system shall be capable of sweeping the temperature range in a 24-hour cycle.

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Primary Subsystem Requirements

- Electrical
  - Standards for both components & connections
  - Properly connecting all subsystems
- Mechanical
  - Materials (surfaces, components, resources) are selected for the application
  - Components operate effectively
  - Properly sized design for easy access, use, and operation
- Software
  - Operating modes
  - User Interface
  - Data Output
  - Control System

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Risk Assessment - Matrix

Consequence	5	B, L	A, E, P	K, M, O	Z	
	4		Y	N	I	Q, W
	3		F, G	T, U, X	V	
	2	R	J		C, D	H
	1	S				
		1	2	3	4	5
		Likelihood				
Notes:	Combined Risk Scores: 0-9 (**Green**) --> Minor 10-19 (**Yellow**) --> Moderate 20-29 (**Red**) --> Major					

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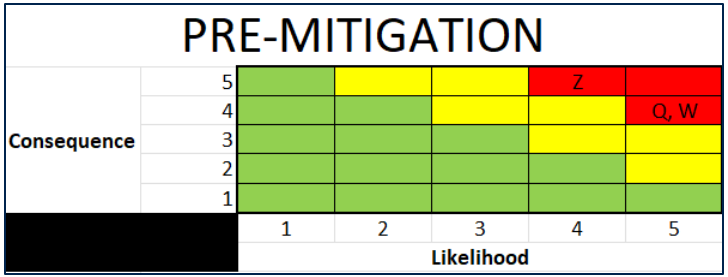
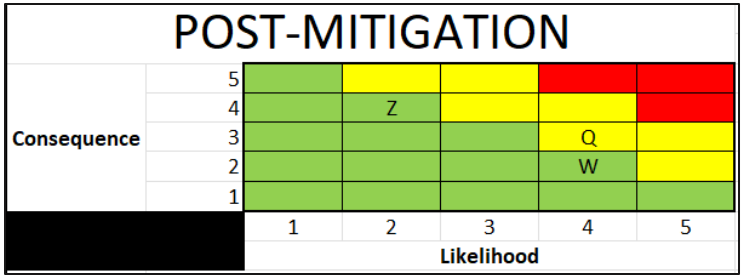
Mitigations

Risk Letter	Risk Score	Description	Effects	Mitigation
Q	20	Software/HMI Failure.	Lack of system input and control.	Hardware Watchdogs, implementing secondary/backup control features.
W	20	Data measuring, collecting, and logging fail to work systematically.	System lagging, inaccurate measurements.	Repeated calibration and testing, and more sophisticated components.
Z	20	Product fails to meet requirements.	Product redesigning, sourcing of new components, falling off schedule.	Early and substantial amounts of research, feasibility testing, and analysis.





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Verification Matrix

Requirement Type	Performance				Functional				Design			Interface				Resource		Verfication Reference
Requirement Title	Thermal range sweep	Outside Skin Temp	Overall Power Draw	Output Data	Door and Human/Machine	Final Weight	Connectivity	Endurance	Load sizing and Energy	Internal Volume	Safety and Lockouts	HMI (Built in)	HMI (Website)	Door function	Interior Assy.	Thermo-Fluid Safety	Overall Power Draw	This column identifies the specific verfication summary related to the activity on each line. Multiple summaries may apply to a single line item.
Level of Assembly																		
Structural																		Includes all structural support, fixtures, and attachments of the system.
Frame						A,T		A	A	A,T					A,D	D		STR-01, STR-02 (frame base dimensions)
Outer Skin		T				A,T				D						D		SAF-02 (temperature)
Handles and Contact Points								A		D								STR-03 (insulation), SAF-02 (temperature)
HMI Cover		M			A	T						A,T						Includes attachment and protection.
Door	D	A,M			T	A,T		A		A,D	A,T			T				CON-04 (thermal control prohibition)
Electrical Component																		Includes all physical electrical components that support the operation and functionalities.
High Voltage			A				A,T	A	A,T								A,T	System Requirement 2.3 (using standard supplied power)
PCB	D		D	A	A		T					A						COM-01 (operatable)
HMI interconnect					A		A,T					T	T					Interface Requirement 4.1 (HMI control)
Sensors	D	M		D			A			I	A,T			D				COM-05 (proper ratings for system operation); includes sufficient room for necessary monitoring/testing components.
Electrical Safety							D	A			T,D				D			COM-02, COM-03, COM-04 (proper rules, equipment, specifications)
Lamps and Outputs	D				T		A											Indicators for operators are tested under all conditions.
Software																		Governs all automatic & manual control of subsystems that support operations.
HMI	D			T	T		A,T					T	T					Manual device attached to system for external control and monitoring. Some functions can be verified independently from other subsystems.
Website	D			T			A						D					Serves for monitoring & ancillary control; verified through testing subsystems.
Controller	D		A	T			A	A,T	D			D		T				CON-1, CON-02, CON-04, CON-05 (control of temperature setpoints and of component when in use)
Thermo-Fluid																		Includes the individual heating and cooling systems and all components that aid in any thermal changes within the chamber.
Assembly (Vacuum Testing)						A										A		Preliminary testing
Layout and Design	T									A	T				I	T		THE-03, THE-04 (containment); no obstruction of different components to structure or functions.
Power Systems	D		T				T	A	A,T							T	T	STR-04 (component insulation)
Design Safety		M									T,I					T		SAF-04, THE-03 (safe-touch from the outside), THE-04 (safe-touch from the outside)

Verification Method:

A- Analysis

D- Demonstration

I- Inspection

M- Measurement

T- Test

V- Validation of Records

Notes:

1. YELLOW: T.
   BLUE: D.
   RED: A.
   GREY: M.
2. Thermo-fluid assembly inlcudes the individual heating and cooling component systems. Assembly testing will need to be proceeded by testing the individual temperature control systems.
3. Each "level of assembly" is a subsystem; the "safety" subsystem is intertwined with the others, and doesn't require it's own level.

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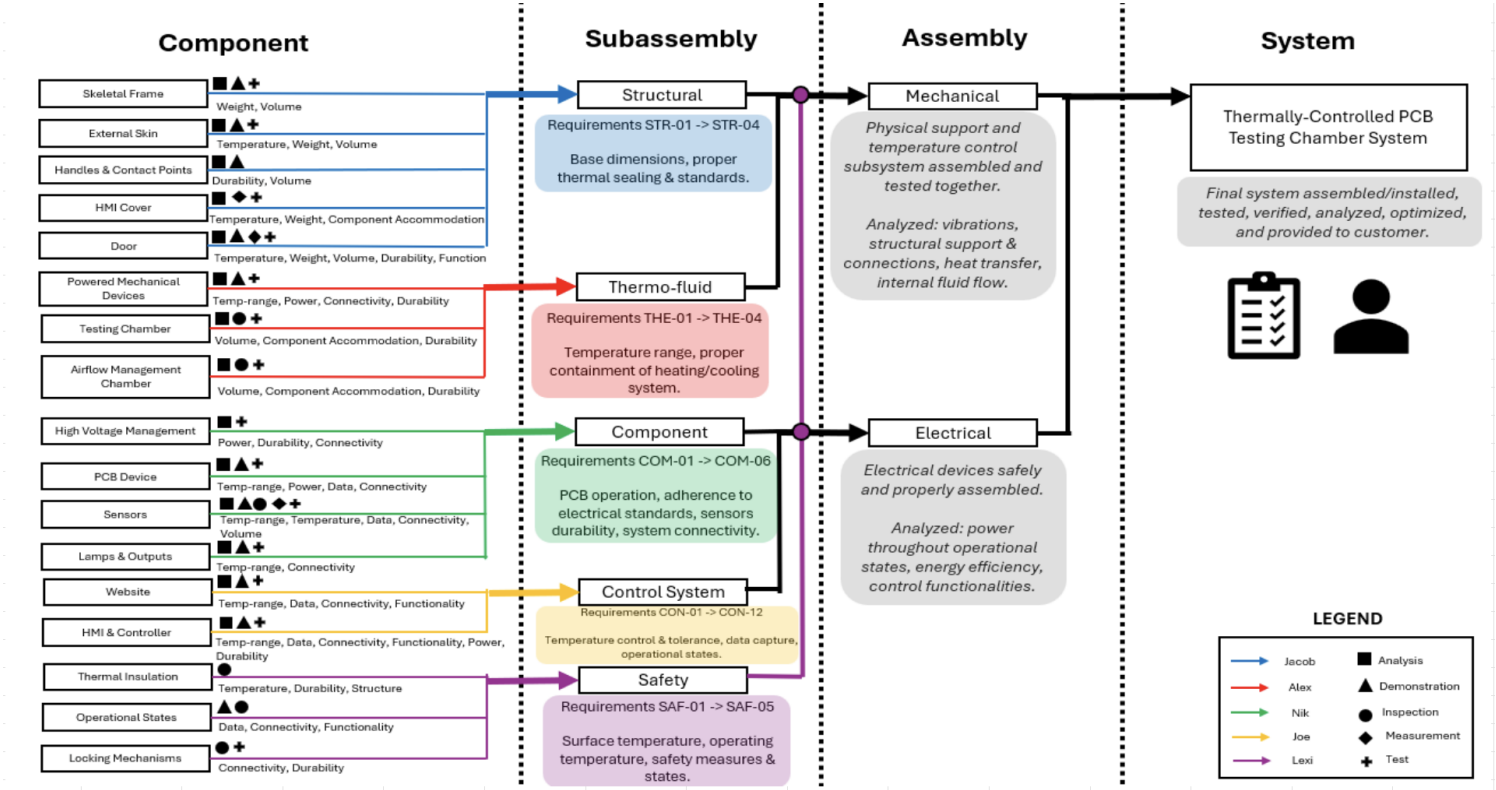
Integration Test & Flow Chart



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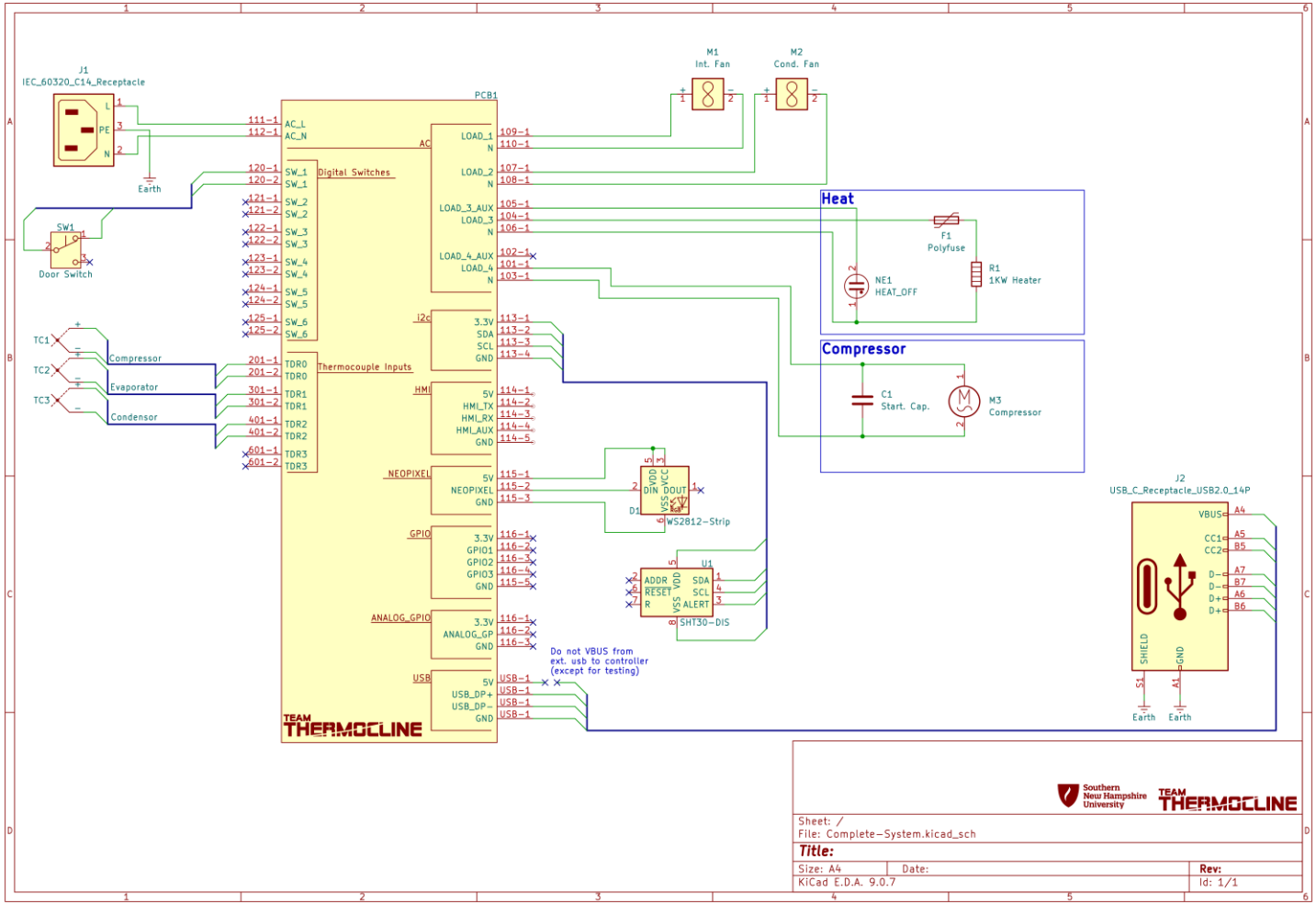
J1
IEC_60320_C14_Receptacle
L
PE
N
Earth
SW1
Door Switch
TC1
TC2
TC3
Compressor
Evaporator
Condensor
PCB1
AC_L
AC_N
AC
Digital Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_6
Thermocouple Inputs
TDR0
TDR1
TDR2
TDR3
LOAD_1
LOAD_2
LOAD_3_AUX
LOAD_3
LOAD_4_AUX
LOAD_4
I2C
3.3V
SDA
SCL
GND
HMI
5V
HMI_TX
HMI_RX
HMI_AUX
NEOPIXEL
GPIO
GPIO1
GPIO2
GPIO3
ANALOG_GPIO
ANALOG_GP
USB
USB_DP+
USB_DP-
TEAM THERMOCLINE
M1
Int. Fan
M2
Cond. Fan
Heat
F1
Polyfuse
R1
1KW Heater
NE1
HEAT_OFF
Compressor
C1
Start. Cap.
M3
Compressor
D1
WS2812-Strip
U1
ADDR
RESET
VDD
VSS
SDA
SCL
ALERT
SHT30-DIS
Do not VBUS from
ext. usb to controller
(except for testing)
J2
USB_C_Receptacle_USB2.0_14P
VBUS
CC1
CC2
D-
D+
SHIELD
GND
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TEAM THERMOCLINE
Sheet: /
File: Complete-System.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1